

Aidan Reaves

970-576-6265 | areaves@mines.edu | aidanreaves.com | github.com/areaves1207 | linkedin.com/in/aidan-reaves

EXPERIENCE

Software Developer & Operations Manager

Jan. 2026 – Present

First Class Motors of Greeley

Greeley, CO

- Sole engineer designing and building **LotManager**, a production full-stack dealership CRM (TypeScript, React, Node.js, PostgreSQL, Docker) replacing a 2007-era Microsoft Access DMS, owning the full lifecycle from requirements analysis to production deployment
- Designed a layered backend architecture (controllers, services, data access layer) with 40+ REST API endpoints, JWT authentication, role-based access control, multi-tenant dealership isolation, and structured audit logging over PII
- Architected and executed a full data migration from a 15-year-old Microsoft Access database (27 tables, 19 views, 250+ column flat schemas) into a normalized PostgreSQL/Supabase schema with 30+ relational tables, proper foreign keys, constraints, and indexing, migrating 4,700+ sales records, 3,800+ customers, 17,800+ payment transactions, and 64,800+ amortization entries through automated Python scripts with data validation
- Integrated a self-hosted LLaMA 3.2 model via Ollama to convert plain-English inventory queries into validated, parameterized SQL filters, scoped per dealership to prevent cross-tenant data access
- Built CI/CD pipelines with GitHub Actions and Dockerized deployment on Ubuntu, including automated unit and integration testing (Jest), Pino structured logging, and Uptime Kuma monitoring and observability

Software Engineering Intern

May 2023 – Aug. 2023

Pie Insurance

Denver, CO

- Built a Python FastAPI CSV validation service to detect schema violations and data inconsistencies in user-submitted documents prior to ingestion, improving reliability and reducing manual validation effort
- Collaborated in an Agile engineering team of six using Git, code reviews, unit testing, and iterative development, partnering with stakeholders to translate business requirements into scalable technical implementations

EDUCATION

Colorado School of Mines

Golden, CO

Master of Science in Computer Science

2025

Colorado School of Mines

Golden, CO

Bachelor of Science in Computer Science

2024

PROJECTS

DreamVault | TypeScript, React, Node.js, Express, MySQL, AWS

Oct. 2025 – Jan. 2026

- Developed and deployed a full-stack SaaS application on AWS (S3, CloudFront, Elastic Beanstalk, RDS), configuring SSL/HTTPS, SPA routing, CORS, and subdomain-scoped session cookies
- Implemented encrypted data storage using AES-256-GCM and searchable encrypted indexing with HMAC tokenization to support full-text search over ciphertext
- Built authentication flows with JWTs, email verification, bcrypt password hashing, protected routes, and secure cookie-based sessions, backed by REST APIs and a modular backend architecture

Paint By Numbers Generator | Python, FastAPI, React, OpenCV, NumPy

Aug. 2025 – Oct. 2025

- Built a full-stack computer vision application that converts uploaded photos into procedurally generated paint-by-number canvases, deployed via Vercel and Render
- Implemented custom image processing algorithms including K-Means clustering, Canny edge detection, connected-component analysis, and automated palette generation
- Optimized backend generation performance by over 80% through algorithm tuning and vectorized array operations

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, C#, Java, C++

Databases: PostgreSQL, MySQL, Microsoft Access

Frameworks/Tools: React, Node.js, Express, FastAPI, Docker, GitHub Actions, AWS, Git, Jest, Tailwind CSS, OpenCV

Concepts: Full-Stack Development, REST APIs, Agile / SDLC, CI/CD, Unit & Integration Testing, Authentication & Authorization, Database Design, Cloud Infrastructure, Monitoring & Observability